

Hands-On: API interaction from React app using Fetch API and Axios

In this lab, you will create a simple React app that interacts with an API using the Fetch API and Axios libraries.

Step 1: Create a new React app

Create a new React app by running the following command in your terminal:

```
npx create-react-app my-app
cd my-app
```

Step 2: Install Axios

Install Axios by running the following command in your terminal:

```
npm install axios
```

Step 3: Create a new component

Create a new file called `App.js` in the `src` directory and add the following code:

```
import React, { useState, useEffect } from 'react';
import axios from 'axios';

function App() {
  const [data, setData] = useState([]);

  useEffect(() => {
    axios
      .get('https://jsonplaceholder.typicode.com/posts')
      .then(response => {
        setData(response.data);
      })
      .catch(error => {
        console.log(error);
      });
  }, []);

  return (
    <div>
      <h1>API Interaction with Axios</h1>
      <ul>
        {data.map(post => (
          <li key={post.id}>{post.title}</li>
        ))}
      </ul>
    </div>
  );
}

export default App;
```

This component uses the Axios library to make a GET request to the JSONPlaceholder API and retrieves a list of posts. The retrieved data is stored in the component's state using the `useState` hook.

Step 4: Test the component

Run the app using **npm start** and test the component by checking if the list of posts is displayed on the page.

Step 5: Implement Fetch API

Create a new file called `FetchApp.js` in the `src` directory and add the following code:

```
import React, { useState, useEffect } from 'react';

function FetchApp() {
  const [data, setData] = useState([]);

  useEffect(() => {
    fetch('https://jsonplaceholder.typicode.com/posts')
      .then(response => response.json())
      .then(data => {
        setData(data);
      })
      .catch(error => {
        console.log(error);
      });
  }, []);

  return (
    <div>
      <h1>API Interaction with Fetch API</h1>
      <ul>
        {data.map(post => (
          <li key={post.id}>{post.title}</li>
        ))}
      </ul>
    </div>
  );
}

export default FetchApp;
```

This component uses the Fetch API to make a GET request to the JSONPlaceholder API and retrieves a list of posts. The retrieved data is stored in the component's state using the `useState` hook.

Step 6: Test the component

Render the `FetchApp` component in the `index.js` file by replacing the `App` component with the `FetchApp` component:

```
import React from 'react';
```

```
import ReactDOM from 'react-dom';  
import FetchApp from './FetchApp';
```

```
ReactDOM.render(<FetchApp />, document.getElementById('root'));
```

Run the app using **`npm start`** and test the component by checking if the list of posts is displayed on the page.

Congratulations! You have successfully implemented API interaction from a React app using Fetch API and Axios.